Tutorial 11

MAT3007 Optimization

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Outline

1 Descent Methods
- A Brief Review on Descent Method
- Convergence and Comparison

2 Projected Gradient Method

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Descent methods

Definition 1 (Descent Direction).

A vector $d \in \mathbb{R}^n$ is a descent direction of f at x if $\nabla f(x)^\top d < 0$.

Descent Method

In $\mathbb{R}^n$, start from an initial point x_0, and generate a sequence of points:

$$x^{k+1} = x^k + \alpha_k d^k$$

Two key points:

Choice for direction:

- Gradient Descent: $d^k = -\nabla f\left(x^k\right)$
- Newton's Method: $d^k = -\left[\nabla^2 f\left(x^k\right)\right]^{-1} \nabla f\left(x^k\right)$

Choice for step size: constant, line search, diminishing,...

Step Size Strategy: Line Search

- Exact line search:

$$\alpha_k = \text{argmin}_{\alpha \geq 0} f\left(x^k + \alpha d^k\right)$$

  Comment: time-consuming

- Backtracking (e.g. Armijo line search): Let $\sigma, \gamma \in (0,1)$, α_k is the largest element in $\{1, \sigma, \sigma^2, \ldots\}$ such that satisfies Armijo condition:

$$f\left(x^k + \alpha_k d^k\right) - f\left(x^k\right) \leq \gamma \alpha_k \nabla f\left(x^k\right)^T d^k$$

  Comment: approximate and cheaper

Exercise 1: Armijo step rule

We consider now the Armijo step rule. Let $U \subset \mathbb{R}^n$ be an open set and $f : U \to \mathbb{R}$. Furthermore, let $\gamma \in (0, 1)$. We have $x \in U$ and $p \in \mathbb{R}^n$ is a descent direction of f at x, then by Armijo rule there exists a $\bar{\alpha} > 0$ such that

$$f(x + \alpha p) - f(x) \leq \gamma \alpha \nabla f(x)^T p, \forall \alpha \in [0, \bar{\alpha}]$$

Please show that the Armijo rule has a feasible solution.

Solution 1

The case $\alpha = 0$ is obviously satisfied. Let $\alpha > 0$ be sufficiently small. We first remark that $x + \alpha p \in U$. Now we calculate:

$$\begin{aligned} & (f(\mathrm{x} + \alpha p) - f(x))/\alpha - \gamma \nabla f(x)^T p \\ \rightarrow \quad & \nabla f(x)^T p - \gamma \nabla f(x)^T p \\ = & (1 - \gamma) \nabla f(x)^T p \end{aligned}$$

Since $\gamma \in (0, 1)$ and $p = -\nabla f(x)$, we have $(1 - \gamma) > 0$ and $\nabla f(x)^T p < 0$. Specifically we can now choose an $\bar{\alpha} > 0$ sufficiently small that for all $\alpha \in [0, \bar{\alpha}]$, it holds:

$$f(x + \alpha p) - f(x) \leq \gamma \alpha \nabla f(x)^T p$$

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Recap: Concepts related to convergence

Theorem 2.

Under some technical conditions on $f(\boldsymbol{x})$, in the gradient method, for all k,

$$f\left(\boldsymbol{x}^{k+1}\right) - f\left(\boldsymbol{x}^*\right) \leq c\left(f\left(\boldsymbol{x}^k\right) - f\left(\boldsymbol{x}^*\right)\right),$$
$$f\left(\boldsymbol{x}^k\right) - f\left(\boldsymbol{x}^*\right) \leq c^k\left(f\left(\boldsymbol{x}^0\right) - f\left(\boldsymbol{x}^*\right)\right)$$

where $\boldsymbol{x}^*$ is a local minimizer, $0 < c < 1$ is some constant.

Corollary 3.

*If $f(\boldsymbol{x})$ is convex, then the gradient method is guaranteed to converge to a **global minimizer**.*

Definition 4 (Linear convergence).

We call the convergence speed specified in the above theorem linear convergence.

Recap: Concepts related to convergence

Definition 5 (quadratic convergence).

Theorem: If $g(x)$ is twice continuously differentiable and x^* is a root of $g(x)$ at which $g'(x^*) \neq 0$, then provided that $\left|x^0 - x^*\right|$ is sufficiently small, the sequence generated by the Newton iterations:

$$x^{k+1} = x^k - \frac{g\left(x^k\right)}{g'\left(x^k\right)}$$

will satisfy

$$\left|x^{k+1} - x^*\right| \leq C\left|x^k - x^*\right|^2$$

with $C = \sup_x \frac{1}{2}\left|\frac{g''(x)}{g'(x)}\right|$.
We call this convergence speed **quadratic convergence**.

Recap: Convergence of Gradient Descent and Newton

Gradient Descent Method

- Globally convergent, doesn't depend on the initial point
- Convergence speed may not be fast enough: Linear convergence

Newton's Method

- quadratic convergence rate
- If the original function is a quadratic function, then Newton's method will converge in just 1 step.

Tradeoff between Newton and Gradient Method

- Gradient method is fast in each iteration and needs less space, however, takes much more iterations to converge
- Newton's method takes much less iterations to converge, however, each iteration is much more expensive in both computation and storage space

Exercise 2: Compare Gradient Descent with Newton's Method

Recall for the Backtracking/Armijo line search, we ensure the following Armijo condition for each iteration:

$$f\left(x^k + \alpha_k d^k\right) - f\left(x^k\right) \leq \gamma \alpha_k \cdot \nabla f\left(x^k\right)^T d^k$$

where α_k is the largest element in $\{1, \sigma, \sigma^2, \ldots\}$. Consider the objective function

$$f(x, y) = 3x^2 + y^4$$

(a) Apply one iteration of gradient descent with $(1, -2)$ as the starting point. The step size shall be choosen by Armijo rule with $\gamma = 0.1, \sigma = 0.5$.

Exercise 2: : Compare Gradient Descent with Newton's Method

Armijo condition:

$$f\left(x^k + \alpha_k d^k\right) - f\left(x^k\right) \leq \gamma \alpha_k \cdot \nabla f\left(x^k\right)^T d^k$$

Objective:

$$f(x, y) = 3x^2 + y^4$$

(b) Set $\gamma = 0.1, \sigma = 0.1$ and start from (1,-2). How does the objective value change?
(c) Apply one iteration of Newton's method with the same starting point and stepsize rule as in (a). How does the objective value of the new iterate compare to that obtained in (a)?

Solution 2

(a). The gradient is $\left[6x, 4y^3\right]^T$, which is $[6, -32]$ at $(1, -2)$. Therefore, $\nabla f(x)^T d = -[6, -32]^T[6, -32] = -1060$ at $(1, -2)$. The Armijo condition can be transformed into

$$f\left(1 - 6\alpha_k, -2 + 32\alpha_k\right) \leq 19 - 106\alpha_k$$

We can iteratively try α_k :

$\alpha_k = 1, \mathrm{LHS} = f(-5, 30)$

$\alpha_k = \frac{1}{2}, \mathrm{LHS} = f(-2, 14)$

$\alpha_k = \frac{1}{4}, \mathrm{LHS} = f\left(-\frac{1}{2}, 6\right)$

$\alpha_k = \frac{1}{8}, \mathrm{LHS} = f\left(\frac{1}{4}, 2\right)$

$\alpha_k = \frac{1}{16}, \mathrm{LHS} = f\left(\frac{5}{8}, 0\right)$

Solution 2

(b). Since the only difference is the σ, we have the same Armijo condition:

$$f\left(1-6\alpha_k, -2+32\alpha_k\right) \leq 19-106\alpha_k$$

We can iteratively try α_k :
$\alpha_k = 1, \mathrm{LHS} = f(-5, 30)$
$\alpha_k = 0.1, \mathrm{LHS} = f(0.4, 1.2)$
Remark: We can see that by choosing a smaller σ, we can obtain a feasible α_k with less iteration. However, the descent on each iteration may be smaller.

Solution 2

(c). The Hessian is

$$H = \begin{bmatrix} 6 & 0 \\ 0 & 12y^2 \end{bmatrix}$$

Therefore, $H|_{(1,-2)} = \begin{bmatrix} 6 & 0 \\ 0 & 48 \end{bmatrix} \Longrightarrow H^{-1}\Big|_{(1,-2)} = \begin{bmatrix} \frac{1}{6} & 0 \\ 0 & \frac{1}{48} \end{bmatrix}$. We have $d^k = H^{-1}\nabla f\left(x^k\right) = \left[-1, \frac{2}{3}\right]^T$. The Armijo condition becomes

$$f\left(1-\alpha_k, -2+\frac{2}{3}\alpha_k\right) \leq 19 - \frac{41}{15}\alpha_k$$

We can show that when $\alpha_k = 1$, LHS $= f\left(0, -\frac{4}{3}\right)$, the Armijo condition is satisfied.

Remark We can see that for one iteration, the Newton's method is not necessarily better than Gradient Descent $\left(f\left(0, -\frac{4}{3}\right) > f\left(\frac{5}{8}, 0\right)\right)$.

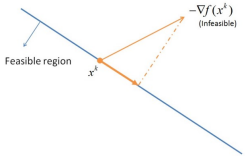
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Recap:Projected Gradient Method

Main idea: Project the descent direction onto the feasible set.



Definition 6 (Definition: Euclidean Projections).

Let $\Omega \subset \mathbb{R}^n$ be a nonempty, closed, convex set. The (Euclidean) projection of $\boldsymbol{x}$ onto Ω is defined as the unique minimizer $\boldsymbol{y}^*$ of the constrained optimization problem:

$$\min_{\boldsymbol{y}} \|\boldsymbol{y} - \boldsymbol{x}\|^2 \quad \text{such that} \quad \boldsymbol{y} \in \Omega$$

and write $\boldsymbol{y}^* = \mathcal{P}_\Omega(\boldsymbol{x})$

Recap: Projection

Theorem 7 (Projection Theorem).

Let Ω be a nonempty, closed, and convex set. Then:

- *A point $\boldsymbol{x}^*$ is the projection of $\boldsymbol{x}$ onto Ω, i.e., $\boldsymbol{x}^* = \mathcal{P}_\Omega(\boldsymbol{x})$, if and only if*

$$(x^* - x)^\top (\boldsymbol{y} - \boldsymbol{x}^*) \geq 0, \quad \forall \boldsymbol{y} \in \Omega.$$

- *The vector $\boldsymbol{x}^*$ is a stationary point of (1) if and only if*

$$\boldsymbol{x}^* - \mathcal{P}_\Omega \left(\boldsymbol{x}^* - \lambda \nabla f\left(\boldsymbol{x}^*\right)\right) = 0 \quad \text{for any } \lambda > 0$$

Recap: Complete Procedure of Projected Gradient Method

- Initialization: Choose an initial point $\boldsymbol{x}^0 \in \Omega$ and $\beta, \gamma \in (0,1)$. For $k = 0, 1, \ldots$:
- Select $\lambda_k > 0$ and compute $\nabla f\left(\boldsymbol{x}^k\right)$ and the new direction $\boldsymbol{d}^k = \mathcal{P}_\Omega\left(\boldsymbol{x}^k - \lambda_k \nabla f\left(\boldsymbol{x}^k\right)\right) - \boldsymbol{x}^k$.
- If $\left\|\boldsymbol{d}^k\right\| \leq \lambda_k \epsilon$, then STOP and $\boldsymbol{x}^k$ is the output.
- Use backtracking line search to find an α^k that satisfies

$$f\left(\boldsymbol{x}^k + \alpha_k \boldsymbol{d}^k\right) - f\left(\boldsymbol{x}^k\right) \leq \gamma \alpha_k \cdot \nabla f\left(\boldsymbol{x}^k\right)^\top \boldsymbol{d}^k$$

  called the Armijo condition.
- Set $\boldsymbol{x}^{k+1} = \boldsymbol{x}^k + \alpha_k \boldsymbol{d}^k$.

Exercise 3: Implement of the Gradient Projection Method

Write a computer code in MATLAB using the gradient projection method to solve the optimization problem (choose starting point to be $(4, 0, 0)$)

$$\begin{aligned} \text{minimize} \quad & e^{x_1+x_2+x_3} + x_1^2 + 2x_2^2 + 3x_3^2 - 2x_1 - 7x_2 - 5x_3 \\ \text{subject to} \quad & x_1 + 2x_2 + 3x_3 = 4 \end{aligned}$$

Also use CVX to solve this problem and compare your solutions.

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